JC2001 Introduction to Software Engineering

Practical Session 2 – Individual Exercises

Software Process and Agile Software Development

Exercise A

The software design paradigm is a part of software development. It includes design, maintenance, and programming. A software paradigm refers to the methods and steps used to design a software system. A software system is a set of interconnecting components that form part of a computer system. It consists of several separate programs, configuration files, etc., which are used to set up these programs. The term "software system" should be distinguished from the terms "computer program" and "software". The term computer program generally refers to a set of instructions (e.g., source code, object code) that perform a specific task. However, a software system generally refers to a broader concept with many more components, such as specifications, test results, end-user documentation, maintenance records, etc.

The Software Development Life Cycle (SDLC) describes all the tasks required to develop and maintain software. It consists of a plan describing how to develop, maintain, replace, and alter the specific software. An SDLC is a process for planning, creating, testing, and maintaining a software system. A well-defined SDLC provides many benefits. The benefits include:

1. Better management control over the entire development project.
2. Transparency and visibility of all the processes.
3. Clarity of requirements.
4. A single version of a product for all the parties involved.
5. Predictable results in terms of time, costs, and deliverables.

Question: You are a project manager at XYZ software house. Your team is responsible for developing a logistics information system for a client. At a project management meeting, the client asks for assurance of product quality and on-time, on-budget delivery of the deliverables. How do you ensure the quality of the product and deliver on the milestones on time and within the budget?

Exercise B

There are various types of SDLC models. These models are referred to as the software development process models. Common SDLC models include:

Waterfall model - It divides project activities into linear sequential phases, where each phase depends on the deliverables of the previous phase and corresponds to a specialisation of tasks. It tends to be one of the less iterative and flexible approaches, as progress flows largely in one direction through the phases of conception, initiation, analysis, design, construction, testing, deployment, and maintenance.

Iterative and incremental development model - The basic idea behind the iterative and incremental development model is to develop a software system through repeated cycles (iterative) and in smaller portions at a time (incremental), allowing software developers to take advantage of what was learned during development of earlier parts or versions of the system. The procedure consists of the initialisation step, the iteration step and the project control list. The initialisation step creates a base version of the system. The goal of the initialisation step is to create a product the user can

interact with. It should sample the key aspects of the problem and provide a solution that is simple enough to understand and implement easily. To guide the iteration process, a project control list is created that contains all tasks that need to be performed. It includes items such as new features to be implemented and areas of redesign of the existing solution. The control list is constantly being revised as a result of the analysis phase.

Spiral model - It is a risk-driven software development process model. Based on the unique risk patterns of a given project, the spiral model guides a team to adopt elements of one or more process models such as waterfall, prototyping or iterative and incremental.

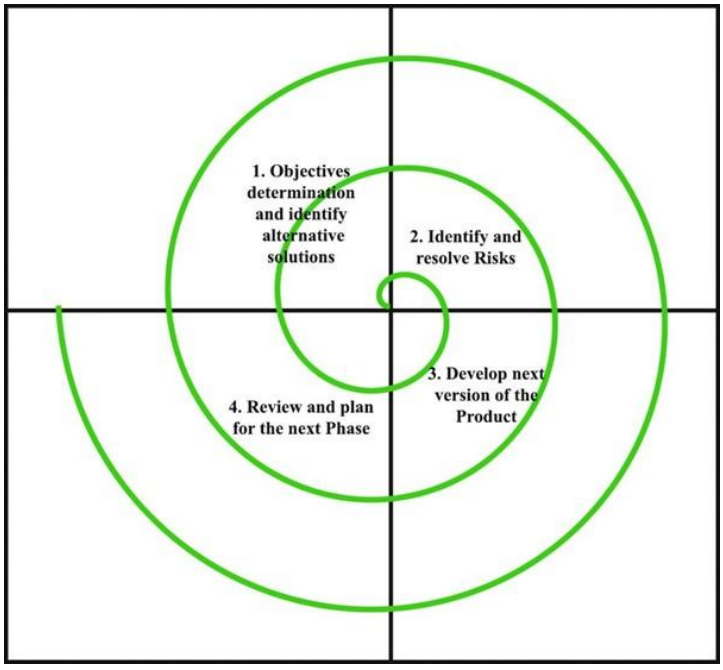


Figure 1: An illustration of the different phases of the Spiral model.

The following article discusses the advantages and disadvantages of five SDLC process models:

1. Vanshika Rastogi, Software Development Life Cycle Models-Comparison, Consequences, *International Journal of Computer Science and Information Technologies*, 2015.

Question: Describe the advantages and disadvantages of the waterfall model, iterative and incremental development model and Spiral model.

Exercise C

A point of sale, or point of purchase, is where you ring up customers. When customers check out, walk up to your checkout counter, or pick out an item from your stand or booth, they are at the point of sale. A point-of-sale (POS) system is the hardware and software that enable your business to make those sales. A POS system allows your business to accept payments from customers and keep track of sales. A POS can work in different ways, depending on whether you sell online, have a physical storefront or both. A POS usually works like this:

1. A customer decides to buy your product or service. If you have a physical store, they may ask a sales assistant to ring them up. The assistant could use a barcode scanner to look up the item's price. For online stores, this step happens when a customer finishes adding items to their cart or basket and clicks the checkout button.

2. Your POS system calculates the price of the item, including any sales tax. Then the system updates the inventory count to show that the item is sold.
3. Your customer pays. To finish their purchase, your customer must use their credit card, debit card, gift card, loyalty points, or cash to complete the payment. Depending on the payment method they choose, your customer's bank then has to authorise the transaction.
4. The point-of-sale transaction is finalised. This is the moment when you officially make a sale. The payment goes through, a digital or print receipt is created, and you ship or hand your customer the items they bought.

Question: Select an SDLC model for developing the POS system and then explain why you selected that model.

Exercise D

Complete the following tasks on agile software development.

1. **Extreme Programming and Scrum**
   a. Use search engines to find explanations of the two topics (Extreme Programming and Scrum). You may also search for the corresponding videos. People who participate in the agile development process in different roles have different views on the agile development process, so please consult various sources.
   b. Try to understand how Extreme Programming and Scrum are used within a software company.
2. **User Stories and Story Cards**
   Suppose you are developing a Student Course Selection System (like what the University uses). Create 5 Story Cards for the system development. You may use the following story card template.

Table 1: Story Card Template

Story Card No: _____	Project Name: _____	Estimation: _____Hours
Story Name:_____		Date: _____
Story:	Acceptance Test:	
Note:	Risk: Low / Middle / High	

References

1. Ian Sommerville, Software Engineering, Pearson, 10th edition, 2015. ISBN-13: 978-0133943030

2. GeeksforGeeks.org, https://www.geeksforgeeks.org/software-paradigm-and-software-development-life-cycle-sdlc
3. Point-of-sale system, https://squareup.com/us/en/townsquare/what-pos-system